

Requirements:

Functional Requirements

1. The management system must support adding new tickets with a unique Ticket ID.
2. When adding a new ticket, students must enter their student number, name, category, priority, status, and description, with strict validation rules (student number must start with "ST").
3. The system must allow ticket searching and filtering to view tickets by student number, ticket ID, status, or priority.
4. The system must generate a ticket summary report showing totals (completed, in progress, open, closed).
5. The system must allow IT staff to update ticket status (Open → In Progress → Resolved → Closed).
6. The system must allow IT staff to add closing comments when resolving/closing a ticket.
7. The system must provide a dashboard displaying ticket counts (open, closed, high priority).
8. The system must validate inputs and reject unsafe data (<script> tags).
9. The system must allow tickets to be archived with confirmation to prevent accidental deletion.
10. The system must provide error messages when invalid input is entered.

Non-Functional Requirements

1. The system must run offline using SQLite (no internet dependency).
2. The system must have a simple, user-friendly interface (Tkinter).
3. The system must respond to search/filter queries in under 2 seconds.
4. The system must ensure data consistency through validation rules.
5. The system must protect student privacy (no unnecessary personal data in reports).
6. The system must be maintainable with modular code and clear documentation.

User Roles

- IT Support Staff: Add, search, filter, update, archive tickets.
- IT Manager: View dashboard, generate reports, monitor performance.
- Students (optional): Submit ticket requests (if extended to web form).

Inputs

- Student number (must start with “ST”)
- Student name (letters/spaces only, not blank)
- Issue category (Login, Network, Hardware, Software)
- Description (10–250 characters)
- Priority (Low, Medium, High, Critical)
- Status (Open, In Progress, Resolved, Closed)
- Closing comment (required when status = Closed)

Outputs

- Ticket list (all tickets with details)
- Search results (matching student number or ID)
- Filtered results (by status/priority)
- Updated ticket status and priority
- Dashboard summary (totals, open, closed, high priority)
- Reports (counts, optional CSV export)

Business Rules

- Student number must start with “ST” and contain digits.
- Student name may not be blank.
- Description must be between 10–250 characters.
- Priority must be one of: Low, Medium, High, Critical.
- Status must be one of: Open, In Progress, Resolved, Closed.
- Closing comment is required when status = Closed.
- Unsafe input (<script>) must be rejected.
- Tickets should be archived rather than permanently deleted.